



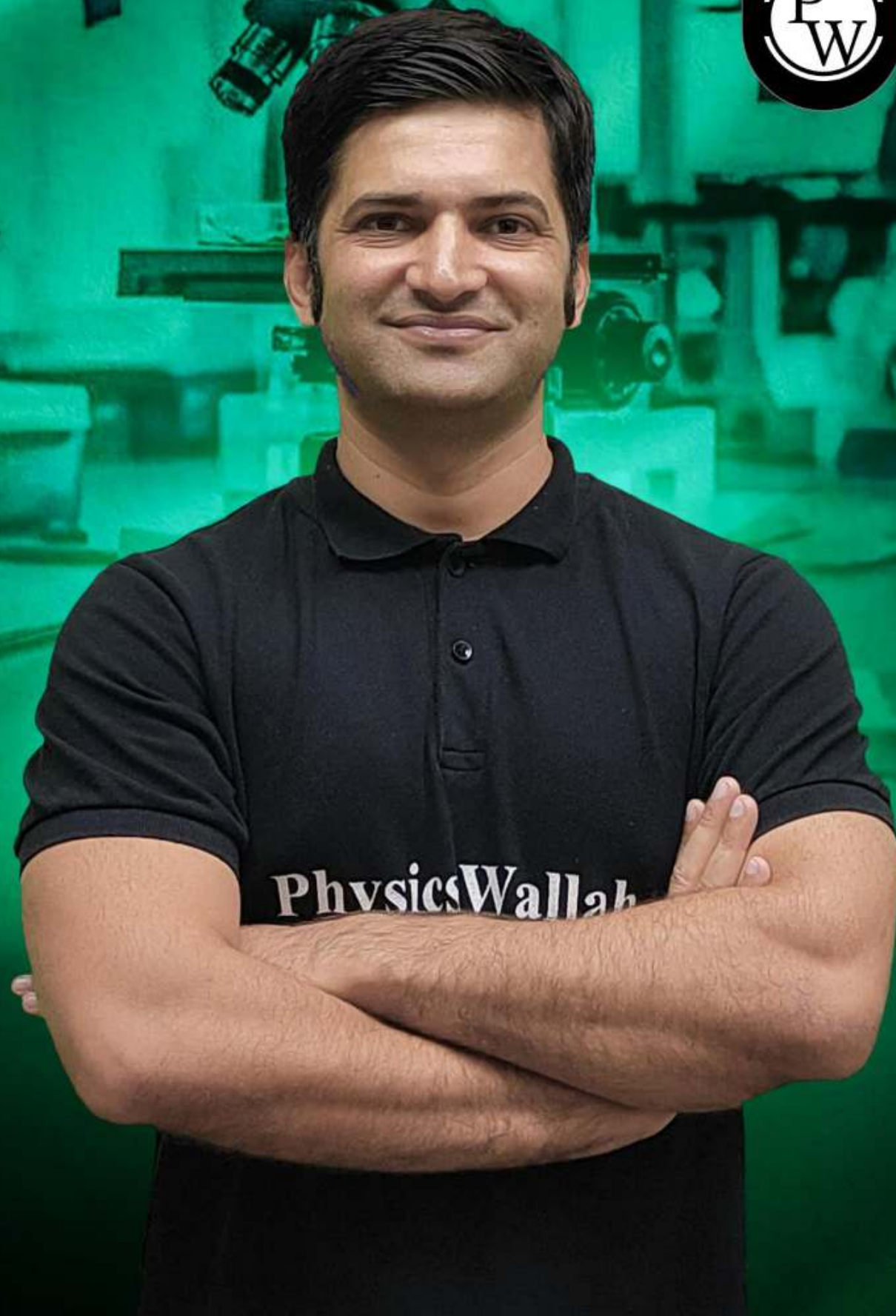
NEET 2.0 2027

Botany

Lecture - 06

Cell : The Unit of Life

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Topics to be covered



1

Cell membrane

2

Endomembrane system

3

ER

4

SHORT NOTES

Cell : The Unit of Life

LEC – 05
BY RUPESH SIR

Composition

Plant ✓

- Cellulose, hemicellulose, pectin, protein

Fungi ✓

- Chitin

Algae ✓

- Cellulose, galactan, mannan, CaCO_3

Primary wall

- Outer, thin
- Continuous
- In meristematic cell (young)
- Capable of growth
- Diminished in old cell/mature

+

Secondary wall

- Inner, thick
- Discontinuous
- Mature/old cell
- Lignin present
- Close to cell membrane

→ **Cell Wall**

Function of Cell wall

- Shape
- Protection from mechanical damage
- Restrict entry of pathogen
- Prevent entry of undesirable macromolecular
- Cell to cell interaction

Middle Lamella

- Common wall b/w two cell
- Calcium pectate (mainly), magnesium pectate

Plasmodesmata

- Cytoplasmic connection b/w two cells
- Cross/traversed by cell wall & middle lamella

Cell Membrane

- 1950, electron microscope, easy to study
- **Chemical study** : RBC membrane **[Lipid + protein]** of human
- **Biochemical study** : Protein + Carbohydrate
- **RBC** : 52% protein + 40% lipid
- This ratio vary in **different membrane**
- **Fluid** (Liquid) **mosaic** (lipid, Protein) model [Singer, Nicolson 1972]
- Common lipid : phospholipid e.g. phosphoglycerides



→ polar/hydrophilic head [outer side] ✓

→ ~~Tail/hydrophilic~~ ^{hydrophobic} [Inner side] :
protection from aqueous / H₂O

• **Bilayer of LIPID present**

• **Cholesterol** : Provide stability to eukaryotic membrane ✓

Two types of Protein [ease of extraction]

External / Peripheral

- 30%
- Easy to remove

Internal / Integral

- 70%
- Difficult

~~Partially buried~~ ^{Buried} : present in single layer of lipid

~~Completely buried~~ ^{Buried} : present in bilayer of lipid

Discussion of Homework of LECTURE – 05



Question

Cell wall

- (A) Non living, rigid covering in fungi & plants ~~only~~
- (B) protect from mechanical damage, infection but ~~not~~ provide shape
- ☒ (C) cell to cell interaction
- (D) barrier to ~~desirable~~ macromolecules

Question

Correct statement

- ~~A.~~ algae cell wall : cellulose, galactan, Mannan ~~not~~ calcium carbonate
 - ☒ B. plant cell wall : protein present
 - ~~C.~~ primary cell wall is capable of growth, ~~inner layer~~
 - ☒ D. as cell mature primary wall diminish
 - ☒ E. secondary wall is formed on inner to primary wall
- (A) 1 (B) 2 ☒ (C) 3 (D) 4

Question

Correct statement

- ☒ (A) Middle lamella consist of calcium pectate which glue different cell
- ~~(B)~~ plasmodesmata connect cytoplasm & it is ~~dead~~ connection
- ☒ (C) cell wall & middle lamella traversed by plasmodesmata
- ☒ (D) both (A) & (C) are correct

8.5.1 Cell Membrane

The detailed structure of the membrane was studied only after the advent of the electron microscope in the 1950s. Meanwhile, chemical studies on the cell membrane, especially in human red blood cells (RBCs), enabled the scientists to deduce the possible structure of plasma membrane.

These studies showed that the cell membrane is mainly composed of lipids and proteins. The major lipids are phospholipids that are arranged in a bilayer. Also, the lipids are arranged within the membrane with the polar head towards the outer sides and the hydrophobic tails towards the inner part. This ensures that the nonpolar tail of saturated hydrocarbons is protected from the aqueous environment (Figure 8.4). In addition to phospholipids, membrane also contains cholesterol.



Question

Correct statement

- (A) Polar head : inner side and hydrophobic tail : outer side
- (B) Major lipid are phospholipid
- (C) Cholesterol absent
- (D) None

Question

Correct statement

- (A) detail structure study after invent of light microscope in 1950
- (B) chemical study on RBC of human help to know possible structure of PM
- (C) cell membrane only composed of lipid and protein
- (D) Lipids are monolayer in cell membrane
- (E) Both (B) and (C)

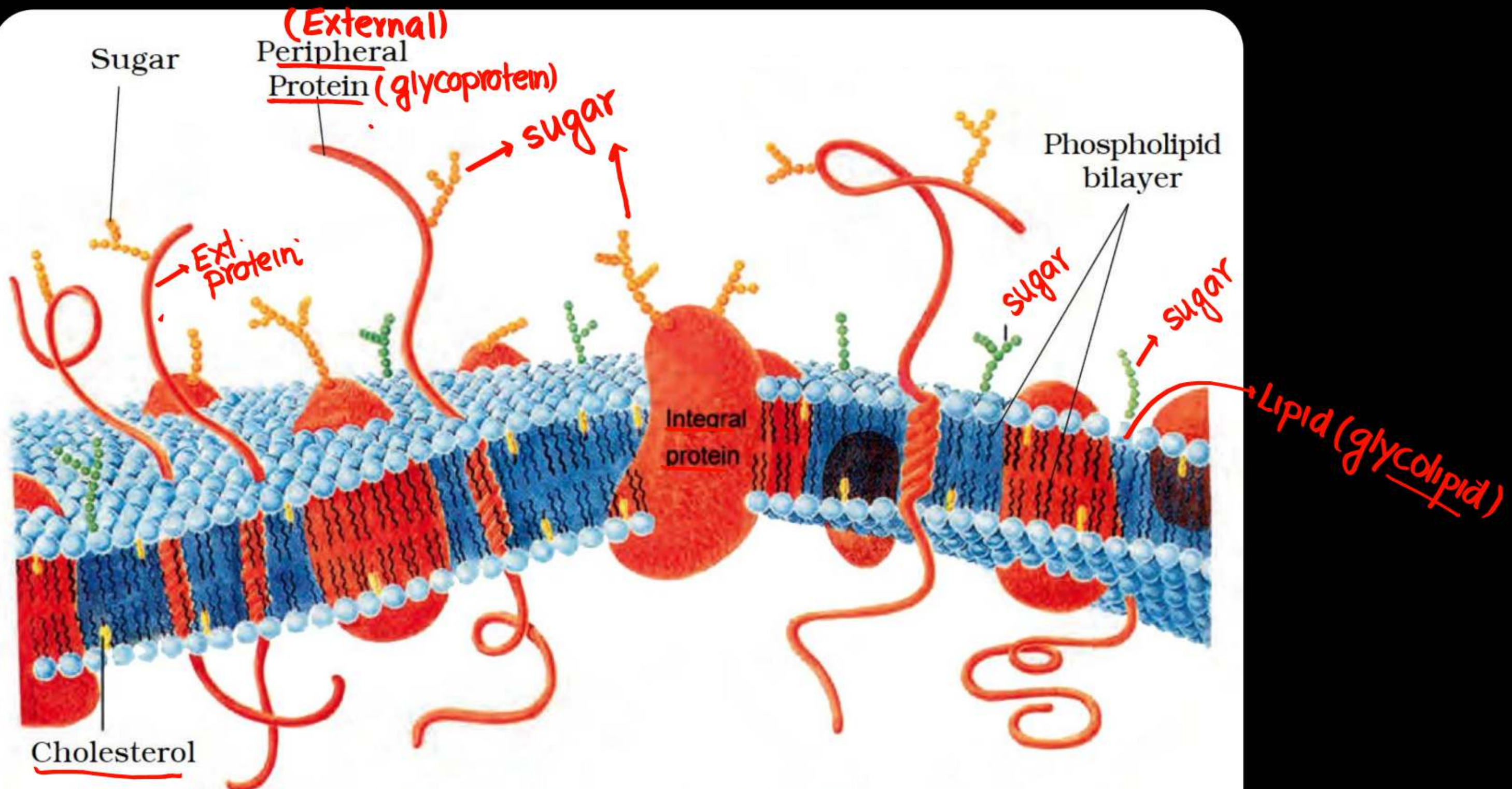


Figure 8.4 Fluid mosaic model of plasma membrane

Later, biochemical investigation clearly revealed that the cell membranes also possess protein and carbohydrate. The ratio of protein and lipid varies considerably in different cell types. In human beings, the membrane of the erythrocyte has approximately 52 per cent protein and 40 per cent lipids.

Question

Assertion : tail towards inner side

Reason : Non polar tail of saturated hydrocarbon protected from aqueous environment

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Question

Correct statement

- (A) biochemical study show membrane also consist of protein only
- (B) Ratio of protein and lipid constant in all cell type membrane
- (C) 52% lipid and 40% protein in RBC of human
- (D) None

STRUCTURE

UPPER LAYER
(outer)

LOWER LAYER
(inner)

glycoprotein

sugar

Cell wall (outer)

glycolipid

polar head / hydrophilic [outer side]
Non polar Tail / hydrophobic [inner side]

Completely
Buried

Partially
Buried

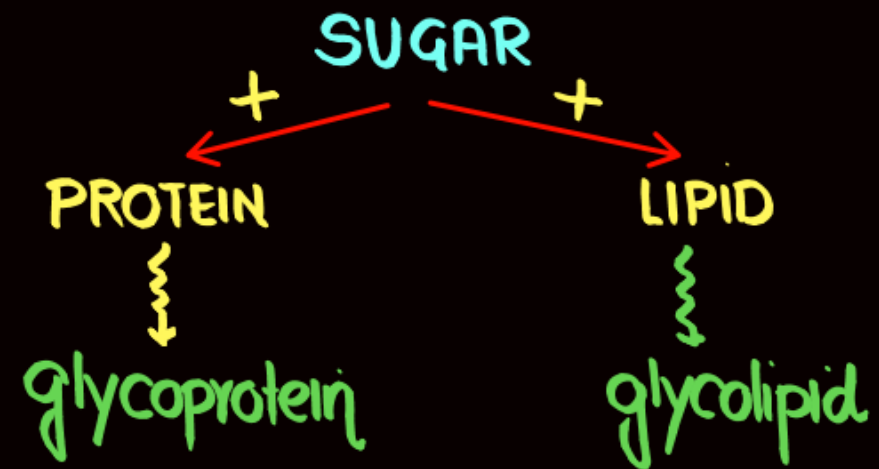
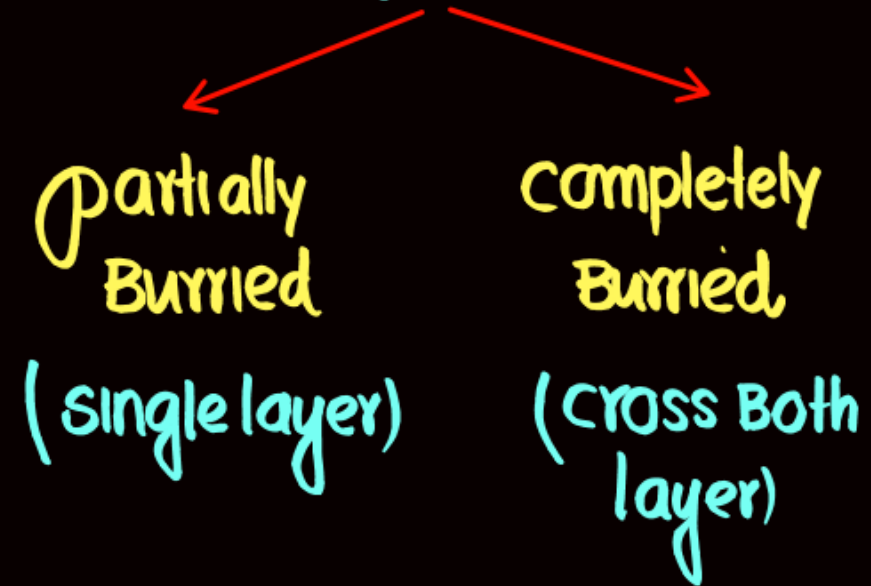
CYTOPLASM (inner)

Cholesterol
(Eukaryote) → provide
stability

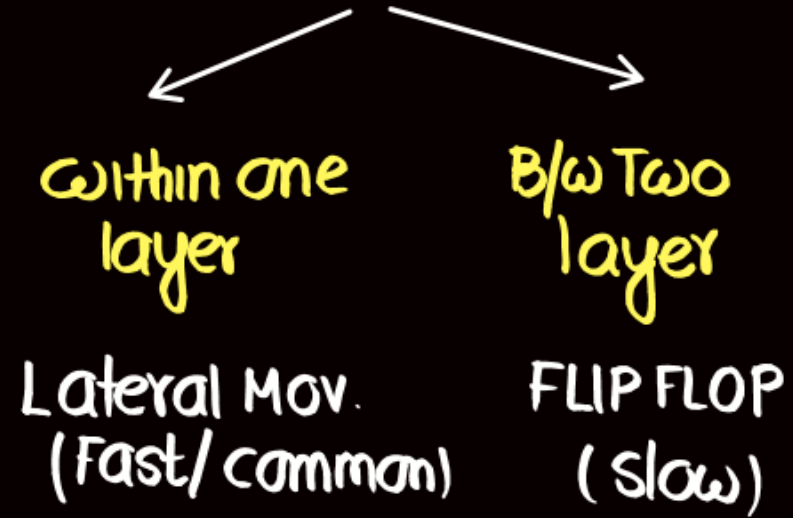
- * Bilayer of LIPID
- * Tail of saturated hydrocarbon protected from (H_2O / aqueous environment) :
so Tail directed inwards

Note:

Integral protein

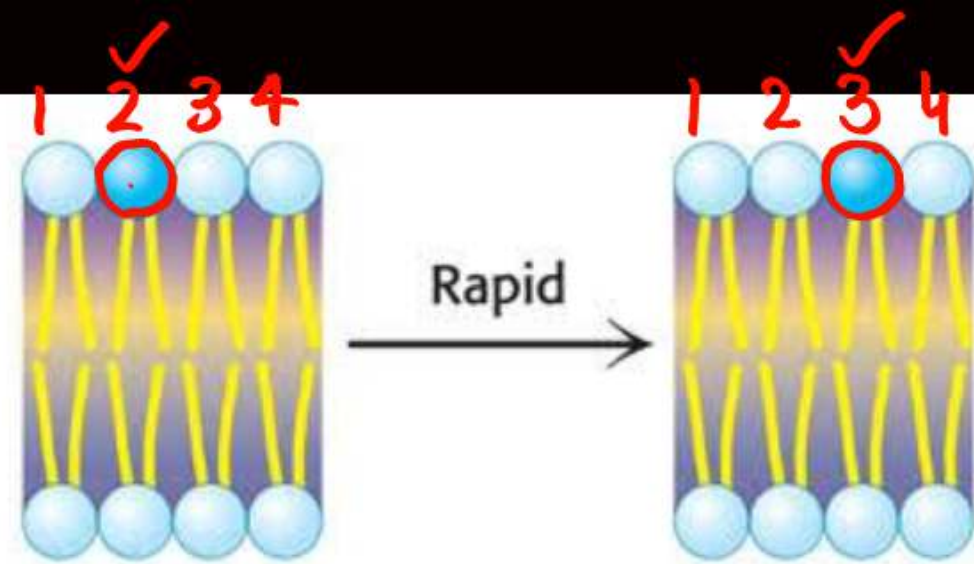


MOVEMENT OF LIPID

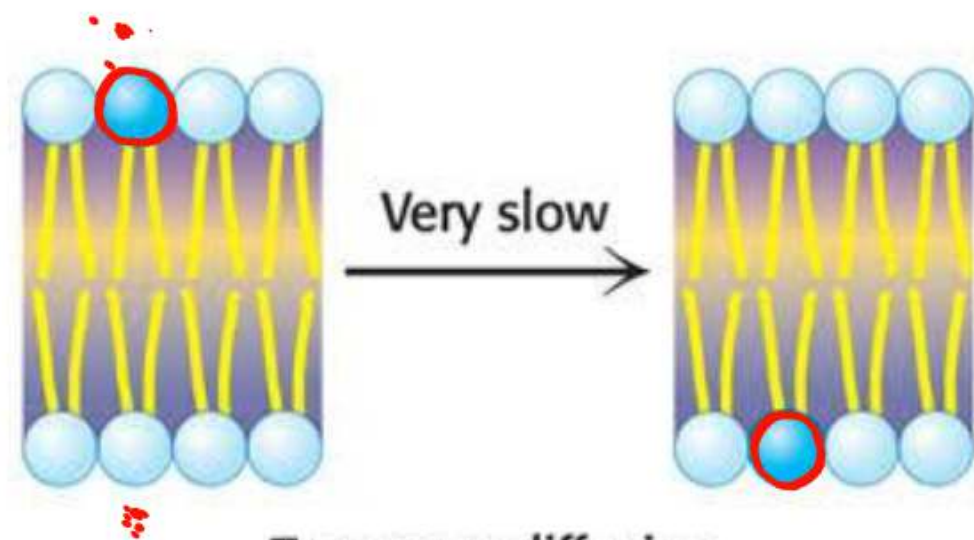


MOVEMENT OF PROTEIN

- ★ NO FLIP FLOP (LARGE SIZE)
- ★ Lateral movement of protein confirm fluid nature of membrane due to lipid



Lateral diffusion



Tranverse diffusion
(flip-flopping)

Function of cell membrane $\Rightarrow$ main: Transport

Simple diffusion

★ movement of molecule from high con^N to low con^N
[along Concentration gradient]

★ downhill

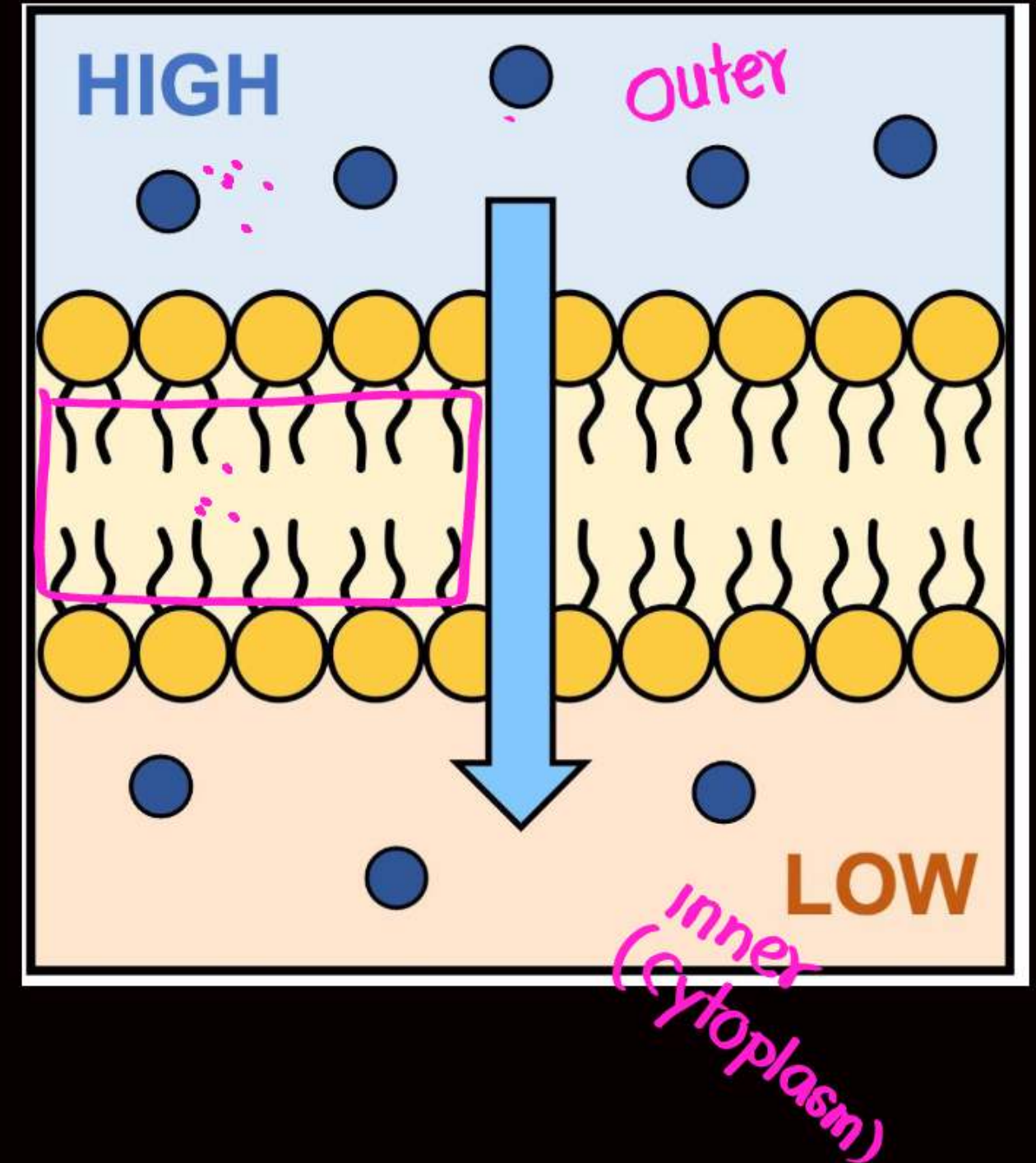
★ No energy [Passive Transport]

★ No involvement of protein

eg: Non polar / Lipid soluble / Uncharged / Neutral molecule
Can pass membrane easily.

★ CO_2 & O_2 gas

★ water (small, polar) can pass at slow Rate.



facilitated diffusion

- ★ High to low (along con^N gradient)
- ★ downhill
- ★ No energy (passive)

Integral protein (Transport)

Carrier

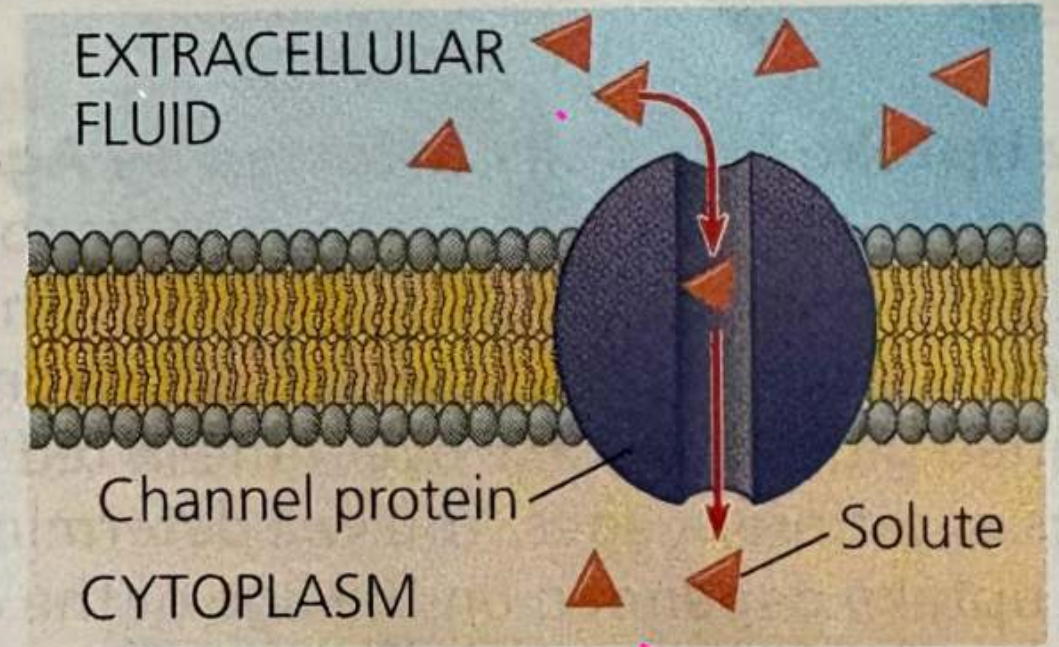
- ★ Molecule Bind
 - ★ protein Rotate
 - ★ molecule Release
- glucose.

- ★ Water soluble/polar molecule difficult to cross Non polar lipid layer so we need facility of carrier protein

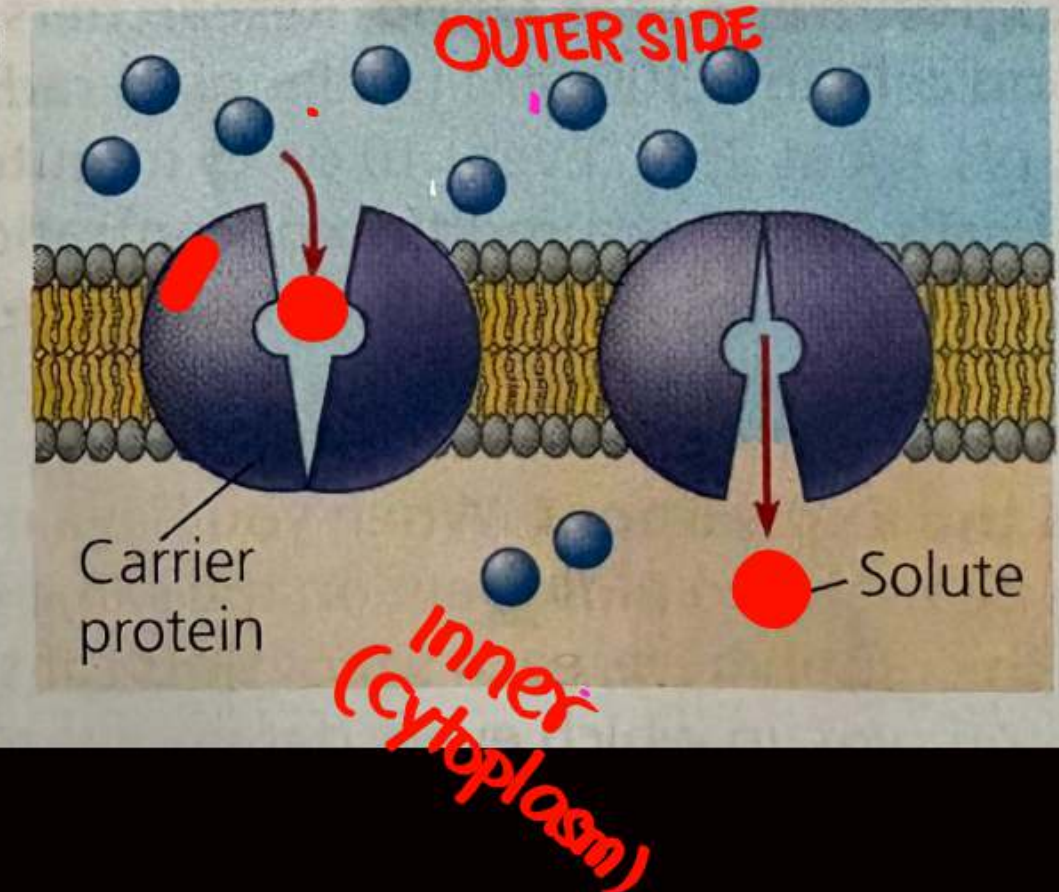
channel/PORINS

- ★ movement of H_2O through channel protein (pore)

- (a) A channel protein has a channel through which water molecules or a specific solute can pass.



- (b) A carrier protein alternates between two shapes, moving a solute across the membrane during the shape change.



Note: most of molecule follow passive Transport

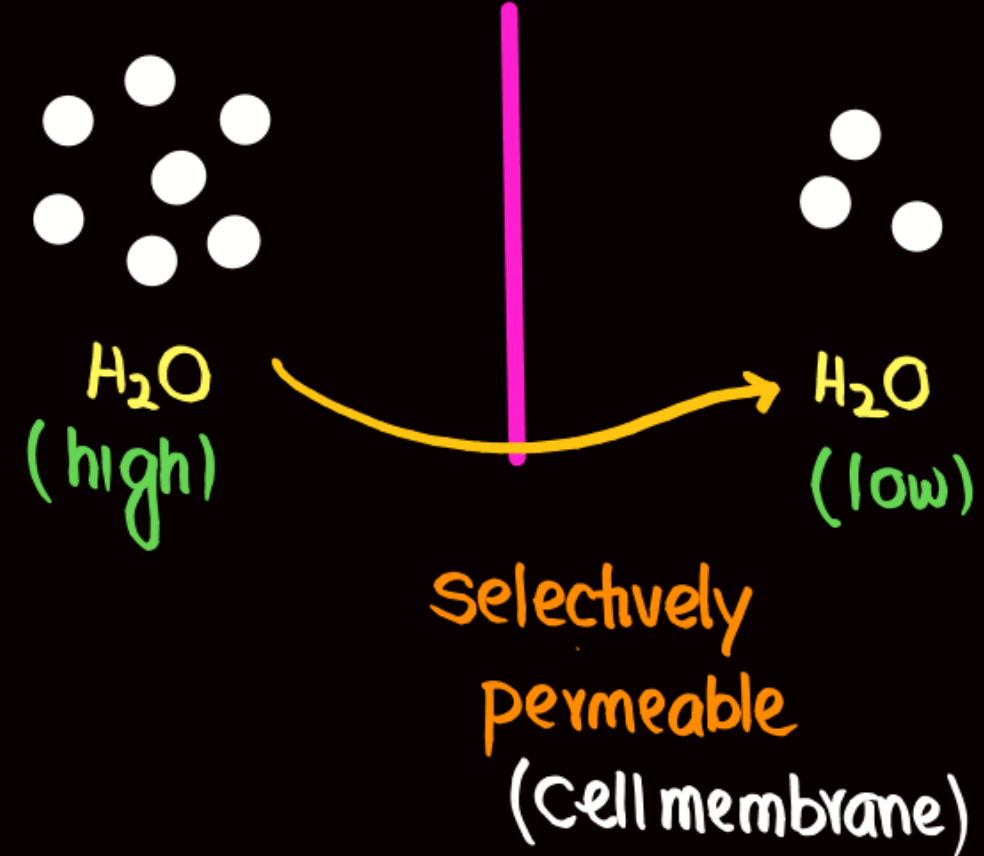
Active Transport

- ★ low to high con^n
- ★ against con^n gradient
- ★ uphill
- ★ need energy
- ★ few ions/molecule follow this Transport
- ★ carrier protein involved.

eg $\text{Na}^+ \text{K}^+$ pump → Protein

Osmosis

- ★ special Type of diffusion



Other function

① endocytosis

FOOD
(SOLID)

enter into
cell

provide
energy to cell
② cell growth

PROTEIN

MOVE
OUTSIDE
CELL
③ SECRETION

④ Intercellular junction
[ZOOLOGY]

⑤ cell division
[Animal cytokinesis]
?

EXTRA

NOTE: : External protein

समझने में है
वही आयेगा

- * sugar of glycoprotein & glycolipid form glycocalyx in animal cell
- * cell to cell Recognition/ Receptor
- * Antigen & antibody RXN

Endomembrane system

inside
cytoplasm

membrane
organelle

WORK IN

COORDINATION

- eg
- ① → ER
 - ② → GB
 - ③ → Lysosome
 - ④ → vacuole
- single layer

Mitochondria

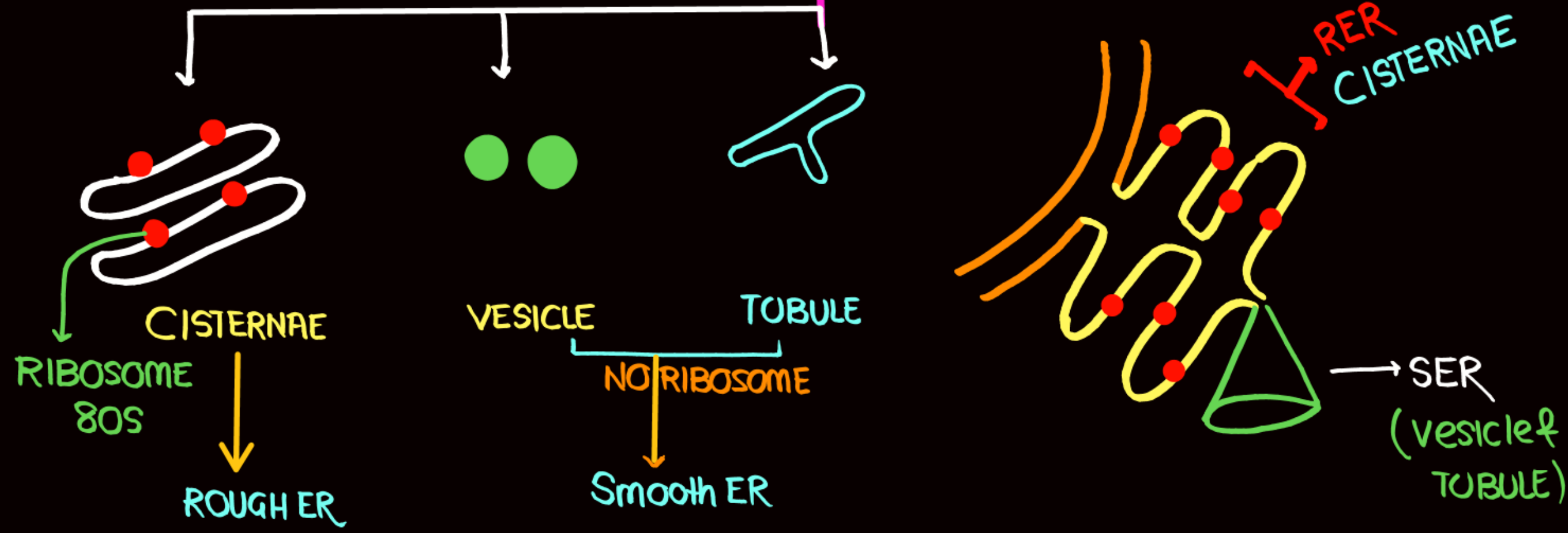
chloroplast

peroxisome

[Non endomembrane
system]

Not work in
coordination
with ① ② ③ ④

Endoplasmic Reticulum



Depending on the ease of extraction, membrane proteins can be classified as integral and peripheral. Peripheral proteins lie on the surface of membrane while the integral proteins are partially or totally buried in the membrane.

[An improved model of the structure of cell membrane was proposed by Singer and Nicolson (1972) widely accepted as **fluid mosaic model** (Figure 8.4). According to this, the ^{semi fluid} quasi-fluid nature of lipid enables lateral movement of proteins within the overall bilayer. This ability to move within the membrane is measured as its fluidity.]

of protein

Question

Correct statement

- (A) Membrane protein are of two type on basis of ease of extraction ✓
- (B) peripheral burried in membrane ✗
- (C) integral protein always completely burried in membrane ✗
- (D) Cell membrane model proposed by singer and Nicolson (1972) as fluid mosaic model ✓
- (E) Both (A) & (D)

The fluid nature of the membrane is also important from the point of view of functions like cell growth, formation of intercellular junctions, secretion, endocytosis, cell division etc. *

Question

Correct statement

- (A) Quasi fluid nature of lipid enable lateral movement of protein
- (B) The ability to move within membrane is measured as fluidity
- (C) Fluid nature help in cell growth, Formation of intercellular junction, secretion, endocytosis, cell division
- (D) All

Question

Correct statement

- ☒ (A) Movement of water by diffusion is Osmosis *most*
- ☒ (B) Only ~~few~~ *most* molecules can pass without energy :passive transport
- ☒ (C) water move from lower to higher concentration in osmosis
- ☒ (D) Few ions move against concentration gradient :Active transport
- ☒ (E) (A) and (D)

Question

How many of the following are correct?

- ☒ a) Sodium-Potassium pumps move across cell membrane by ~~passive transport~~
- ☒ b) The quasi-fluid nature of lipid enables lateral movement of proteins within lipid bilayer
- ☒ c) Polar solutes require carrier protein to facilitate their transport across the membrane
- ☒ d) Both ~~facilitated~~ diffusion and active transport are energy dependent processes
- ☒ e) Neutral solutes may move across the membrane by simple diffusion.

- (A) 1 (B) 3 ☒ (C) 4 (D) 2

One of the most important functions of the plasma membrane is the transport of the molecules across it. The membrane is selectively permeable to some molecules present on either side of it. Many molecules can move briefly across the membrane without any requirement of energy and this is called the **passive transport**. Neutral solutes may move across the membrane by the process of simple diffusion along the concentration gradient, i.e., from higher concentration to the lower. Water may also move across this membrane from higher to lower concentration. Movement of water by diffusion is called **osmosis**. As the polar molecules cannot pass through the nonpolar lipid bilayer, they require a carrier protein of the membrane to facilitate their transport across the membrane. A few ions or molecules are transported across the membrane against their concentration gradient, i.e., from lower to the higher concentration. Such a transport is an energy dependent process, in which ATP is utilised and is called **active transport**, e.g., Na^+/K^+ Pump.



Homework From **Yakeen NEET Module**



Exercise – 1 (Topic wise)

Q. 34, 36, 37, 38, 39, 40, 41, 42, 43
44, 45, 46, 47, 48, 49



Thank

You